transient metal layer at the interface between the ablation plume and water environment followed by its decomposition into large molten nanoparticles, (2) nucleation, growth, and rapid cooling/solidification of small nanoparticles at the very front of the emerging cavitation bubble, above the transient interfacial metal layer, and (3) spinodal decomposition of a part of the ablation plume located below the transient interfacial layer, leading to the formation of a large population of nanoparticles growing in a high-temperature environment through inter-particle collisions and coalescence. The atomistic simulations start with a certain energy density distribution in the target material and do not consider the plasma formation dynamics in the target and the surrounding liquid. Therefore, they are most valuable for the regime in which energy deposition occurs through IBA by electrons in the metal without ionization and without a contribution of liquid breakdown. Furthermore, because of their numerical expense, such simulations are restricted to selected parameters. A systematic exploration of the parameter dependence of the entire sequence of LAL processes is still lacking. The $R_{max}(E_L)$ curves in Figs. 5 and 6 resulting from such systematic exploration helped to delineate the regimes of scenarios 1 to 3 for plasma formation in bulk liquid. Measurement of $R_{max}(E_L)$ for the approximately hemispherical bubbles produced at solid targets may help to delineate the regimes and transitions between scenarios I and II in LAL. Reich et al. already showed that scaling laws for LAL ablation rates and bubble size are related [92], and this approach can be further sophisticated through a larger and more detailed data base.

An energy balance of LAL and LIBS processes requires knowledge of the energy partitioning into phase transitions and the mechanical pathways of shock wave emission and bubble formation. The $R_{max}(E_L)$ data yield the bubble energy, and plasma pressure and shock wave energy can be assessed by optical measurements of the initial shock wave propagation [40,114,228]. The shock speed $u_S(t)$ derived from measured $r_S(t)$ data together with the Hugoniot relations linking shock speed and pressure then yields $p_s(r)$ [40,330]. Since most shock wave energy is dissipated already in close plasma vicinity, its approximate amount can be obtained by integrating over the energy dissipation losses $\varepsilon_{SW}(r)$, which is accessible by evaluation of $p_s(r)$ [114]. The energy remaining in the far-field can be obtained by measuring and evaluating the far-field pressure profile [229], and the sum of both contributions yields the total energy [114]. This approach together with $R_{max}(E_L)$ data provides important elements for the creation of an *energy balance for LAL and LIBS processes*. The balance can be completed if the absorbed (non-reflected) fraction of the incident laser energy and its partitioning into vaporization energies of target and liquid are also known. Such data may come from measurements of the ablated target volume and photographs with high spatiotemporal resolution from which the vaporized liquid volume can be estimated.

Accepted Manuscript

Optimization strategies for LAL suggested by the findings of the present study: For optimum LAL and LIBS performance the plasma energy density should remain at a moderate level to maximize ablation efficiency and avoid shielding of the target by plasma formation in the liquid in front of the target. The use of laser pulses with reproducible shape (slm pulses) largely improves the reproducibility of ns optical breakdown and will likely facilitate to establish a clear gap between breakdown threshold at the target and in the liquid to minimize shielding and support a mono-disperse particle size distribution. This is due to the increased sharpness and the higher breakdown threshold with slm pulses compared to mlm pulses (Table 2, Figs. 4 and 5). Furthermore, ns pulses at IR wavelengths provide a better separation of $F_{\mathrm{th,liquid}}$ from $F_{\mathrm{th,target}}$ than shorter wavelengths because their breakdown threshold in water is higher due to the larger hurdle for multiphoton-induced seed electron generation with smaller photon energies. Use of a top hat beam instead of a Gaussian beam consolidates the separation of $F_{\mathrm{th,liquid}}$ from $F_{\mathrm{th,target}}$ because in a Gaussian beam, liquid breakdown may be ignited in the center, while in its periphery plasma is formed only at the target. Spatially multimode beams often have an approximately top hat profile. Otherwise, the beam profile of a Gaussian beam can be somewhat homogenized by insertion of a helical phase plate that transforms it into a propagation-invariant vortex beam with ring-shaped cross section [309]. This approach avoids hot spots in the liquid in front of the target that may lead to unwanted plasma formation in the liquid. Finally, the laser beam should be focused behind the target to avoid an intensity peak in the liquid, and the spot size should be adjusted to provide radiant exposure values around 2.7 times ablation threshold, which yield maximum ablation efficiency in blow-off processes [75,188].

Target shielding during pulse series in LAL. In LAL, maximization of the ablated nanoparticle mass and the optimization of their size distribution and reactive properties are a major goal. Therefore, pulse series with large average power are often employed [73,338], which leads to the problem that bubble remnants and bubbles formed at nanoparticles shield the target during consecutive ablation events [91,93,336,339]. Shielding by bubble remnants can be avoided by lateral pulse separation through fast scanning [73,338], by pulse sequences, which favorably influence the bubble dynamics to clear the ablation site [316,340], and by flow configurations [341,342]. For envisioning strategies to minimize bubble formation at nanoparticles, one needs to look at the mechanisms of bubble formation, which depend on the laser wavelength used for ablation. In metallic nanoparticles, light-driven collective oscillations of free electrons induced by electromagnetic waves form plasmons. Plasmon oscillations heat the particle, and they enhance the external electric field through charge concentration at the polar points, i.e. along the polarization direction of the field [343,344]. Near fields are evanescent, and the field-enhanced region is thin [10-20 nm for a gold NP with 50 nm radius, as shown in Fig. 15(a)].

Accepted Manuscript

Absorption and heating peak at resonance wavelengths in the visible wavelength range, where the frequency of incident light matches with that of the plasmons, and drop off rapidly for longer wavelengths [Fig. 15(b)]. The field enhancement peaks at the resonance wavelength but is still larger than 10-fold at IR wavelengths [Fig. 15(c)]. As a consequence, bubble formation near resonance is driven by heat diffusion from the heated nanoparticle into the surrounding liquid, while off-resonance plasma formation may occur in the liquid surrounding the particle, if the laser irradiance in the field-enhanced regions exceeds the breakdown threshold [345-347]. This results in a more vigorous effect driven by a volumetric phase transition of the liquid in the plasma even though the particle temperature is lower.

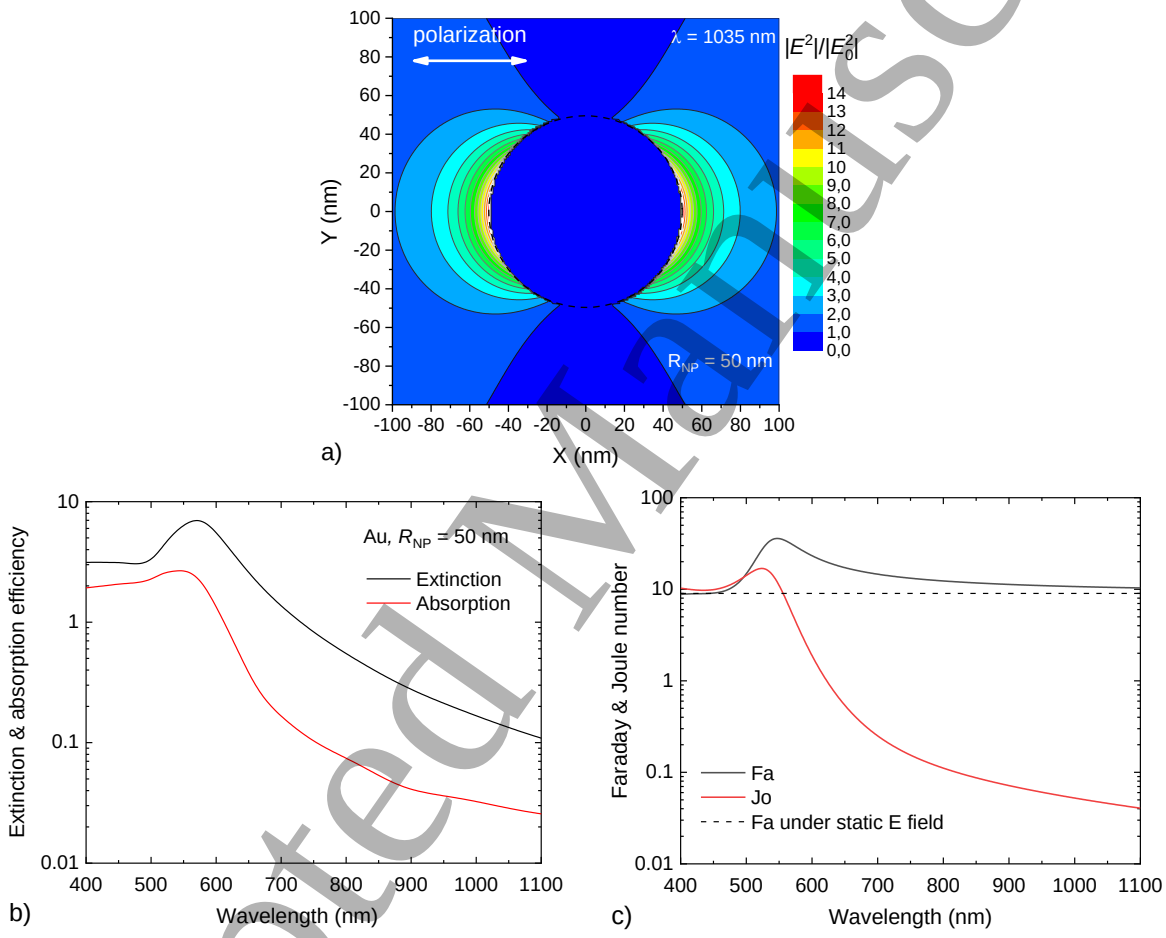

Figure 15 Wavelength dependence of nanoparticle properties illustrated on a gold nanoparticle with 50 nm radius. (a) Near-field enhancement around the nanoparticle at 1035 nm wavelength, calculated using the Matlab tool box *SPlaC v1.0* from Ref. [344]. For this purpose, the original tool box enabling 2-D calculations was extended to allow for 3-D calculations. Maximum enhancement of field intensities is achieved along the *x*-axis in the polarization direction of the external field. (b) Wavelength dependence of plasmonic absorption and extinction (absorption + scattering) in a gold nanoparticle of 50 nm radius, calculated using Ref. [348]. (c) Wavelength dependence of Faraday number (Fa) and Joule number (Jo) characterizing the near-field enhancement outside the nanoparticle and the ability of the electric field inside a metallic nanoparticle to generate heat, respectively [349,350]. The Faraday number is the maximum value of the near-field enhancement factor outside the NP, and the Joule number characterizes the ability of a metal to generate heat in an electric field.

Accepted Manuscript